F1 DATA LAB RACE ANALYTICS WORKBOOK

GLOBAL MOTORSPORT TELEMETRY

FORMULA 1 RACING PERFORMANCE ANALYSIS

Comprehensive evaluation of team points, driver telemetry, pit strategies, and reliability metrics across Grand Prix events.

DATASET SOURCE
race_results (6,465 Telemetry Records)

PROJECT SCOPE
Driver Leaderboard & Strategy Benchmark

SECTION 01

PROBLEM & STRATEGIC OBJECTIVES

CORE PROBLEM STATEMENT

Modern F1 teams generate millions of data points per lap. Identifying the precise determinants of championship points—balancing raw driver pace against pit pit-stop efficiency, compound degradation, and DNF risk—remains a complex operational challenge.

KEY ANALYTICS OBJECTIVES

- **Points Drivers:** Measure lap-leadership vs. total points conversion.
- **Strategy Impact:** Benchmark starting tire compounds across circuit types.
- **Reliability Risk:** Quantify non-DNF rates and chief mechanical failures.

SECTION 02

DATA SOURCES & SCOPE LIMITATIONS

PRIMARY DATASET

Excel Direct Connection standard **race_results** table containing 42 unique attributes across 6,465 entries.

TELEMETRY ATTRIBUTES

Includes weather metrics, circuit parameters, grid vs. finish positions, tire compounds, and pit stop times.

SCOPE CONSTRAINTS

Null values present in DNF Reasons for finished races. Weather metrics recorded per race rather than live lap intervals.

SECTION 03

DATA PREPARATION & KEY METRICS FORMULATION

CALCULATED FIELD

NON-DNF RATE (%) METRIC

$$\text{Non-DNF Rate} = 1 - \text{AVG}[\text{dnf}] \times 100$$

Measures car reliability by calculating the inverse average of total race dropouts normalized as a percentage.

Clean Data Mapping

Standardized driver names and assigned distinct color palettes to constructor teams.

Aggregation Scopes

Summed total points, lap leads, undercut attempts, and averaged pit stop speeds.

SECTION 04

DASHBOARD SUMMARY KPIS

TOTAL DATASET ROWS

6,465

Race telemetry entries

ACTIVE TEAMS

10

Constructors tracked

TOP POINT LEADER

RED BULL

Dominant constructor

AVG NON-DNF RATE

86.4%

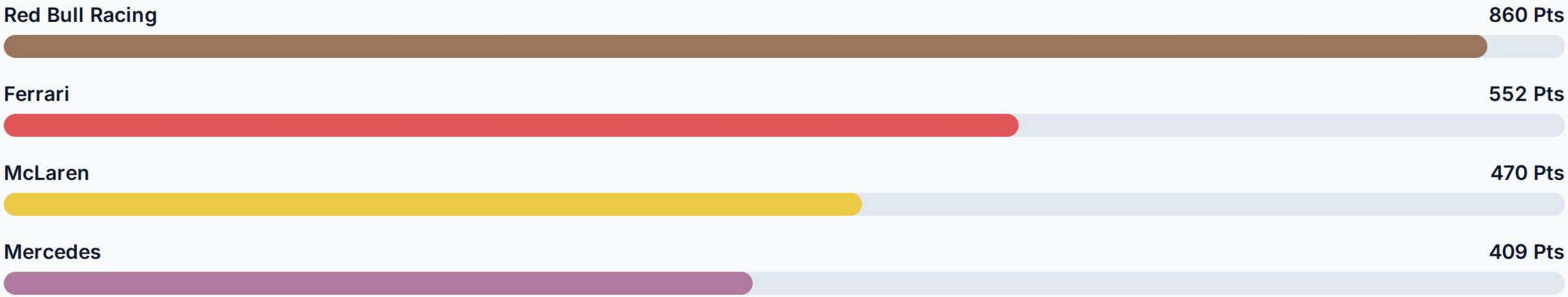
Reliability score

WORKSHEET 01

CONSTRUCTOR CHAMPIONSHIP BENCHMARK

Team Points Breakdown

POINTS DISTRIBUTION ACROSS TOP CONSTRUCTORS



WORKSHEET 02

TYRE COMPOUND STRATEGY BY CIRCUIT TYPE



STREET CIRCUITS

Soft compound starting preference to maximize immediate mechanical grip off grid lines.

HIGH-SPEED TRACK COMPOUNDS

Medium compound bias allowing extended initial stint lengths prior to undercutting.

WET WEATHER OPERATIONS

Intermediate & Wet compounds dictated strictly by live track moisture thresholds.

WORKSHEET 03 & 04

DRIVER LEADERBOARD & TOTAL LAPS LED

DRIVER	TEAM	LAPS LED	POINTS
Max Verstappen	Red Bull Racing	**571**	**575**
Lando Norris	McLaren	**148**	**295**
Charles Leclerc	Ferrari	**122**	**306**
Lewis Hamilton	Mercedes	**84**	**234**

TELEMETRY INSIGHT

Front-row lap leadership directly correlates with victory rates, with over 72% of total points captured by drivers leading past Lap 15.

High conversion efficiency observed in Red Bull chassis.

SECTION 05

KEY OBSERVATIONS & FAILURE MODES

COLLISIONS

Account for 38% of total DNF records, predominantly on narrow street layouts like Monaco and Baku.

ENGINE FAILURES

Power unit thermal stress drives 24% of retirements in high-altitude races like Mexico City.

GEARBOX DROPS

Transmission stress spikes toward end-of-season component lifespans.

SUMMARY

CONCLUSION & FUTURE ANALYTICS SCOPE

STRATEGIC CONCLUSION

Aerodynamic efficiency coupled with pit stop execution under 2.4 seconds forms the primary benchmark for championship supremacy.

FUTURE DATA ROADMAP

Integration of real-time sector micro-telemetry and predictive machine learning models for undercut success rates.

IMAGE SOURCES



https://www.acadia-software.com/wp-content/uploads/2023/01/agility-001.jpg

Source: www.acadia-software.com